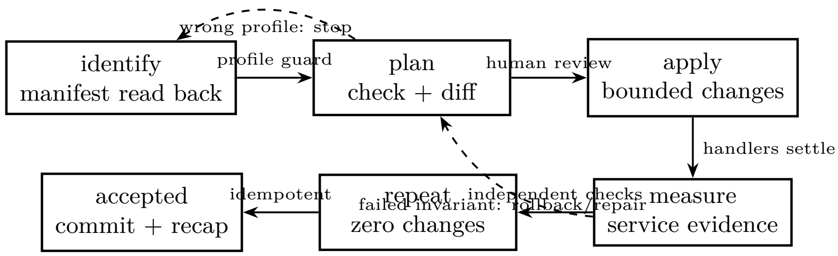
Convergence Design

High-level design

What day two looks like, and why installation is deliberately small

Installation does only what cannot be done later: partition, encrypt, write boot artifacts, pin one interface, create one account. Everything else is a repository, a commit and a converge run. A machine is not a thing that was set up once; it is a thing that is continuously made to match what is written down.

A run is a measured state transition



A GREEN TASK RECAP IS NOT ACCEPTANCE

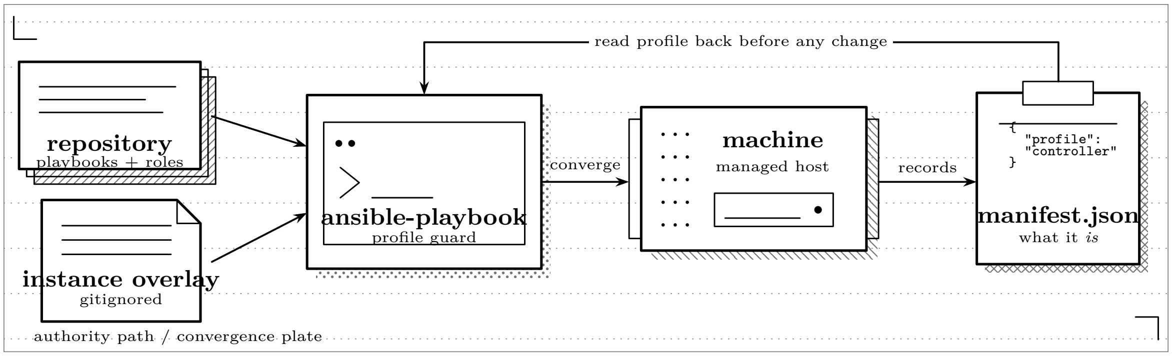
Ansible reports whether modules completed and whether they claim to have changed state. Acceptance also needs observations from the host and a client: bound sockets, active units, generated-file digests, lease behavior, route absence and a second zero-change plan. The playbook cannot be its own only witness.

Where the line falls

Install time	**Why it cannot wait**
Partitioning and LUKS2	Destructive, needs the authorization gate, and cannot be re-run on a machine in use. GOVERNED BY ADR 0051, ADR 0058
Boot artifacts and the UKI	The machine must boot before anything can converge it.
The managed interface pin	A MAC-pinned `.link` file has to exist before first boot decides whether to serve DHCP. GOVERNED BY ADR 0050
The manifest	Records what the machine *is*. Everything afterwards reads it rather than being told. GOVERNED BY ADR 0060
Converge time	**Why it belongs there**
Packages, accounts, sshd	Changes constantly. A reinstall to add a package is a design failure.
dnsmasq, nginx, iPXE config	Generated from the manifest by the same code the installer uses, so a decision has one implementation.
Optional services	Off by default, enabled per instance, movable to another host. GOVERNED BY ADR 0054
Directory membership	Enabled only once the directory exists and answers. GOVERNED BY ADR 0055

The manifest is the authority

Convergence never tells a machine what it is. It asks.



The profile guard	Each playbook reads `/etc/homelab/manifest.json` and refuses to proceed if the profile is not the one it converges. Converging a Workstation with the Controller playbook would start DHCP on it — the one thing forbidden at every stage. GOVERNED BY ADR 0008
Not inventory	Inventory says how to *reach* a machine. What the machine is comes from the machine. An inventory typo can therefore fail to connect, but cannot convert a Workstation into a DHCP server.
Disposable is recorded	A development-proof installation says so in its manifest, and convergence says so out loud. That is a recorded state, not a remembered one. GOVERNED BY ADR 0043, ADR 0060

One implementation of each decision

The Controller's `dnsmasq`, `nginx` and iPXE configuration is not an Ansible template. It is produced by the same generators the installer and the bootstrap host call, invoked through a small bridge and installed verbatim.

Why this is worth the indirection

A template is a second implementation. The generators refuse configurations that violate a recorded decision — a Controller address inside its own DHCP pool, a default route advertised, `bind-dynamic` where `bind-interfaces` is required — and those refusals are unit-tested. A template that drifted would still be valid `dnsmasq` syntax, would pass `dnsmasq --test`, and would quietly stop enforcing the decision it was written to enforce. The tests assert byte equality between what convergence installs and what the generator produces, so the drift cannot happen silently.

Evidence travels with the change

A convergence record binds intent, implementation and observation. At minimum it names the repository commit, inventory target, manifest profile, first-run recap and second check-mode recap. For generated files it also records a digest from the generator side and one read back from the host.

Command or source	Independent observation	Acceptance
`cat /etc/homelab/manifest.json`	parsed profile and disposable flag	Exact intended profile before any role changes state.
check-mode playbook	reviewed task/file diff	No unexplained deletion, restart or secret-bearing value.
generator output	host-side `sha256sum`	Byte-identical configuration, not merely valid syntax.
`systemctl is-active`	`ss -lntup` and client query	Required units active and bound only where designed.
second check run	recap with changed count	Zero changed and zero failed.

Answer before continuing: Does the manifest read-back name the intended profile? Does the reviewed diff correspond to the reviewed commit? Which external observation will prove each restarted network service is safe and reachable?

Failure branches are designed states

Observed failure	Immediate state	Next evidence
Profile mismatch	No changes; play ends failed	Capture manifest and inventory target; correct identity, not the guard.
Generator refusal	Existing installed file remains	Save rejected inputs and generator diagnostic; correct the network plan.
Config syntax failure	Service not restarted	Preserve candidate and syntax output; fix generation or role logic.
Activation invariant fails	DHCP and HTTP both stopped	Read first-boot journal; identify interface, address or competing DHCP authority.
Second run changes	Not accepted	Name the changing task and before/after value; repair idempotence before another mutation.
Directory unavailable	Local break-glass path remains	Prove local key login at console/LAN before attempting join or auth changes.

Getting back in when the directory is down

The directory is a bootstrap dependency. While it is unavailable, only cached logins and one local account work.

CONVERGENCE REFUSES TO STRAND A MACHINE

Both the base role and the directory-join role assert that a break-glass administrator key is configured, and stop if it is not. It is a dedicated key pair used for nothing else: the daily-driver key is loaded in an agent all day and forwarded to other hosts, so reusing it would protect the recovery account with the credential most likely to be part of a bad day. Nothing in the repository ever handles the private half. GOVERNED BY ADR 0055, ADR 0063

The domain join itself is deliberately not automated. It requires directory-administrator credentials, and automating it would mean storing those somewhere a playbook can read unattended. The role detects that a machine is not joined, stops, and prints the one command a person runs at a console.

Public repository, private overlay

In Git — Playbooks, roles, generators, tests, decisions, and a template overlay in which every value is a visible placeholder.

Not in Git — Hostnames, addresses, interface MACs, disk serials, DHCP reservations and the break-glass public key. These live in a gitignored overlay that needs its own backup, because nothing else is backing it up. GOVERNED BY ADR 0046

Not anywhere — The LUKS2 passphrase, private keys, directory-administrator credentials and Kerberos keytabs. The overlay is gitignored, not encrypted, and sits in a working tree that gets copied around.

Enforced, not intended — The site build scans published sources for address literals and internal hostnames and fails closed. The template is tested to contain no key material. A tracked template that drifts from the variables the roles read is a test failure.

What is proven and what is not

Stage	State	What it establishes
firmware	passing	A lab guest boots UEFI on its serial console, finds no boot disk, and attempts PXE over IPv4 on a segment with no route off it.
boot-chain	pending	The Controller answers DHCP, serves iPXE over TFTP, and the guest chainloads the installer over HTTP.
install	pending	The genuine installer is driven to completion over the guest's serial console, serial typed and all.
activation	pending	First boot refuses on a segment that already has DHCP, and starts cleanly on one that does not.
converge	pending	Ansible converges the installed Controller, and a second run reports no changes.

Acceptance measurements

`sha256sum <generated-file>`	Matches the digest computed from the reviewed generator output for this commit and overlay.
`systemctl is-active dnsmasq nginx`	Both active only after activation preflight; partial activation is failure.
`ss -lntup`	Listeners match the intended managed address and ports; no wildcard exposure appears unexpectedly.
`ip route show default`	On an isolated client, no output; the Controller did not advertise a nonexistent router.
`ansible-playbook ...  --check`	Recap reports zero changed and zero failed after the applying run.

The retained record is small: date/time, operator, reviewed commit, target manifest digest, generated-file digests, both recaps, client lease and route observations, and any explicitly pending matrix stage. Secrets and private-overlay contents are referenced by identifier, never copied into it.

PENDING IS REPORTED, NEVER SKIPPED

A stage that cannot run yet prints what it is waiting for and the matrix does not call itself green. A harness that silently skips what it cannot do converts an unknown into a false assurance, which is worse than having no harness at all.

A defect this design found before hardware did

The lab's virtual disk reported no serial. The installer will not offer a disk whose serial cannot be typed at the authorization prompt, so the acceptance matrix could never have installed anything — it would have failed with "no eligible disk" on its first real run, and the obvious diagnosis would have been that the installer was broken. The fix is one line of QEMU argv, but the reason it was found is that the fixtures describe hardware *shapes* rather than one captured machine: NVMe, SATA and eMMC partition naming, absent serials, removable media, wireless-only. A snapshot of one box would have missed it.